

Appendix B: Notation Reference

Appendix B: Notation Reference I

This appendix collects all notation, symbols, and technical terminology used throughout the manuscript. Entries are grouped by domain and ordered alphabetically within each section. The **First Use** column indicates the section where each term is first introduced or defined.

Sections A–K: A Linguistic Terms; B Category Theory; C Enriched Categories and Magnitude; D Distributional Semantics and LLMs; E Distributional Active Inference (DAIF); F Active Inference and Cognitive Models; G Quantum and Topological Terms; H Logical and Type-Theoretic Terms; I AI and Communication Protocols; J Diagrammatic Reasoning; K Notation Conventions.

Appendix B: Notation Reference II

Figures that colour-code case roles (category graphs, string diagrams, appendix panels) take those colours from the shared map `CASE_COLORS` in `src/visualization/styles.py` (keys are `CaseRole` names). Captions state the mapping where it matters for reading the diagram.

Linguistic Terms I

Table 1: Linguistic terminology and symbols.

Term	Symbol	Definition	First Use
Ablative	ABL	Case role encoding origin, source, or cause	§2
Accusative	ACC	Case role encoding patient, theme, or direct object	§2
Active–Stative alignment	—	Alignment in which the sole argument S splits by agentivity	§2
Agent-like argument	A	The agent-like argument of a transitive clause	§2
Alignment functor	$F : \mathcal{U} \rightarrow \mathcal{L}$	Structure-preserving map from universal to language-specific case category	§2

Linguistic Terms II

Alignment type	—	Systematic pattern governing how S, A, P are grouped for case marking	§2
Case category	$\mathcal{C}$	Small category whose objects are case roles and morphisms are grammatical relations	§2
Case frame	—	The set of case roles activated by a particular verb or predicate	§2
Case-typed noun space	$N_{\text{NOM}, \text{ACC}}$	Case-specific vector subspace in a case-enriched DisCoCat model	§4
Categorial grammar	—	Grammar assigning each lexical item an algebraic type encoding combinatory potential	§3
Contraction	$a^l \cdot a \rightarrow 1$	Pregroup reduction eliminating an adjoint pair	§3

Linguistic Terms III

Dative	DAT	Case role encoding recipient, goal, or beneficiary	§2
Deep case	—	Fillmore's universal semantic primitive (e.g., Agentive, Objective)	§2
Ergative–Absolutive alignment	—	Alignment grouping $S = P \neq A$	§2
Expansion	$1 \rightarrow a \cdot a^l$	Pregroup expansion introducing an adjoint pair	§3
Fluid-S	—	Alignment in which S marking varies by context or volition	§2
Fluid-S functor	F_θ	Context-dependent alignment functor parameterized by a volition feature θ	§2
Functors (Alignment)	$F_{\text{acc}}, F_{\text{erg}}$	Alignment functors $\mathcal{U} \rightarrow \mathcal{L}_{\text{acc}}$ resp. $\mathcal{U} \rightarrow \mathcal{L}_{\text{erg}}$	§2

Linguistic Terms IV

Language-specific case category	$\mathcal{L}_{acc}, \mathcal{L}_{erg}$	Codomain categories for accusative vs. ergative alignment functors	§2
Genitive	GEN	Case role encoding possessor or source	§2
Formal semantics	—	Montague's programme: meaning assigned compositionally via truth-conditional functions	§4
Grammatical relation	—	Morphism in a case category relating two case roles	§2
Instrumental	INS	Case role encoding instrument or means	§2
Kāraka	—	Pāṇini's system of deep semantic roles in Sanskrit grammar	§2
Left adjoint	a^l	Left adjoint type satisfying $a^l \cdot a \leq 1$	§3
Locative	LOC	Case role encoding location or context	§2
Markedness	—	Asymmetry in the formal complexity of paradigmatic oppositions	§2

Linguistic Terms V

Monadic semantics	—	Song's extension using monads to model sublexical verb-root decomposition	§3
NominativeNOM		Case role encoding agent, experiencer, or intransitive subject	§2
Nominative— Accusative align- ment	—	Alignment grouping $S = A \neq P$	§2
Passivization	—	Syntactic transformation promoting patient to subject position	§3
Patient- like argu- ment	P	The patient-like argument of a transitive clause	§2
Pregroup gram- mar	—	Grammar where types have left and right adjoints forming a compact closed category	§3

Linguistic Terms VI

Proto-Agent	—	Dowty's cluster of agentive entailments (volition, sentience, causation)	§2
Proto-Patient	—	Dowty's cluster of patient entailments (change of state, causal affectedness)	§2
Right adjoint	a^r	Right adjoint type satisfying $a \cdot a^r \leq 1$	§3
Sole argument	S	The sole argument of an intransitive clause	§2
Thematic role	—	Semantic relation between a predicate and its argument (Agent, Patient, Goal, etc.)	§2
Tripartite alignment	—	Alignment in which $S \neq A \neq P$ (all three distinguished)	§2
Verb root decomposition	—	Analysis of a verb's internal causative and result-state layers via a monad	§3

Linguistic Terms VII

Vocative VOC Case role encoding direct address

§2

Category Theory I

Table 2: Category theory terminology and symbols.

Term	Symbol	Definition	First Use
Box	—	Node in a string diagram representing a morphism	§3
Cap	$\eta : 1 \rightarrow a^r \otimes a$	Unit of a compact closure (coevaluation map)	§3
Category	$\mathcal{C}$	Collection of objects and morphisms with identity and associative composition	§2
Classifying topos	$\mathcal{E}_{\mathbb{T}}$	Canonical topos: models of $\mathbb{T}$ in a Grothendieck topos $\mathcal{F}$ correspond to geometric morphisms $\mathcal{F} \rightarrow \mathcal{E}_{\mathbb{T}}$	§6
Codomain	$\text{cod}(f)$	Target object of a morphism f	§2

Category Theory II

Commutative diagram	—	Diagram in which all directed paths with the same start and end yield equal composites	§1
Cobordism	—	Manifold with boundary connecting two lower-dimensional manifolds; domain of TQFT functors	§8
Compact closed category	—	Monoidal category in which every object has a left and right dual	§3
Composition	$g \circ f$	Sequential application of morphisms: first f , then g	§2
Cup	$\varepsilon : a \otimes a^r \rightarrow 1$	Counit of a compact closure (evaluation map)	§3
Diagram	D	DisCoPy representation of a morphism in a monoidal category	§3

Category Theory III

†-compact closed category	—	Compact closed category with a contravariant involutive endofunctor $\dagger$; framework for ZX-calculus	§8
Domain	$\text{dom}(f)$	Source object of a morphism f	§2
Fiber bundle	—	Projection $\pi : E \rightarrow B$ whose fibers carry role-filler structure; topos-theoretic LoT model	§6
Functor	$F : \mathcal{C} \rightarrow \mathcal{D}$	Structure-preserving map between categories	§2
Geometric theory	$\mathbb{T}$	Theory axiomatized by sequents with finite conjunctions, arbitrary disjunctions, existential quantification	§6
Geometric theories (case)	$\mathbb{T}_{\text{typ}}, \mathbb{T}_{\text{log}}, \mathbb{T}_{\text{dis}}, \mathbb{T}_{\text{enr}}$	Typological, type-logical, distributional (DisCoCat), and enriched case theories (??)	§6

Category Theory IV

Bridge topos	$\mathcal{E}_{\mathbb{T}_{\text{bridge}}}$ (schematic)	Intermediate classifying topos linking two formalizations in the Morita bridge programme	§6
Identity mor- phism	id_A	Morphism from A to itself satisfying $f \circ \text{id} = f = \text{id} \circ f$	§2
Monad	—	Endofunctor with unit and multiplication satisfying associativity and unit laws	§3
Monoidal category	—	Category with a bifunctor $\otimes$ (tensor product) and unit object I	§3
Morita equiva- lence	$\mathcal{E}_{\mathbb{T}_1} \simeq \mathcal{E}_{\mathbb{T}_2}$	Equivalence of classifying toposes enabling inter-theoretic transfer	§6
Morphism	$f : A \rightarrow B$	Arrow between objects in a category; encodes a relation or transformation	§2
Natural transfor- mation	$\alpha : F \Rightarrow G$	Family of morphisms $\alpha_A : F(A) \rightarrow G(A)$ commuting with all morphisms; alignment maps in semantics use $\nu, \dots, \nu' : \dots$	§2

Category Theory V

Object	$A, B,$	Entities in a category; in our framework, case roles	§2
Presheaf	$\hat{\mathcal{C}} = [\mathcal{C}^{op}, \mathbf{Set}]$	Contravariant functor from $\mathcal{C}$ to Set	§6
Snake equation	$(1 \otimes \varepsilon) \circ (\eta \otimes 1) = 1$	Zigzag identity: fundamental axiom of compact closed categories	§4
String diagram	---	Planar graph faithfully representing morphisms in a monoidal category	§3
Subobject classifier	Ω	Object in a topos playing the role of a truth-value object	§6
Swap	$\sigma_{A,B} : A \otimes B \rightarrow B \otimes A$	Braiding morphism permuting two objects in a symmetric monoidal category	§3

Category Theory VI

Tensor product	$A \otimes B$	Monoidal product representing parallel composition or concatenation	§3
Topos	$\mathcal{E}$	Category with products, exponentials, and a subobject classifier; generalized universe of sets	§6
Universal construction	—	Object or morphism characterized by a universal property (product, coproduct, limit)	§6
Wire	—	Edge in a string diagram representing a type (object)	§3

Enriched Categories and Magnitude I

Table 3: Enriched categories and magnitude terminology.

Term	Symbol	Definition	First Use
Base-change functor	—	Conceptual tower $\mathbf{Bool} \hookrightarrow [0, 1] \hookrightarrow \mathbf{R}_{\geq 0}$ showing progressively richer enrichments of grammatical categories (not computationally instantiated; this paper implements only $[0, 1]$ -enrichment)	§3
Categorical magnitude	$ X $	Sum $\sum_i w_i$ where $(w_1, \dots, w_n)$ solves $Z\mathbf{w} = \mathbf{1}$; measures effective number of objects	§5
Composition inequality	$\mathcal{V}(A, B)$	Enriched analogue of composition on $\mathcal{V}(B, C)$; $\mathcal{V}(A, C)$: composite hom-value is at least the product of intermediates	§5
Enriched category	$\mathcal{V}$ -Cat	Category whose hom-sets are replaced by objects of a monoidal category $\mathcal{V}$	§5

Enriched Categories and Magnitude II

Enriched functor	—	Structure-preserving map between enriched categories respecting hom-values	§5
Hom-value	$\mathcal{V}(A, B)$	Enriched analogue of hom-set; measures degree of relation between objects	§5
Identity axiom	$\mathcal{V}(A, A)$	Every object has maximal self-relatedness in the enriched hom (equality, not mere inequality, in our $[0, 1]$ -convention)	§5
Similarity matrix	$Z_{ij} = \mathcal{V}(A_i, A_j)$	Matrix of all pairwise hom-values; used to compute categorical magnitude	§5
Lawvere metric space	—	Category enriched over $([0, \infty], +, 0)$; generalizes metric spaces via enriched categories	§5
Magnitude homology	$MH_n(\mathcal{C})$	Graded homological invariant categorifying magnitude; detects higher-dimensional holes in enriched categories	§5
Morphism weight (precision)	w_f, w_k, w_c	Enriched weights in $[0, 1]$; subscript denotes morphism or role in VMP/DPE (??). Convention: w_f always refers to the <i>enriched</i> hom-value $\mathcal{P}(A, B)$ in the	§7

Enriched Categories and Magnitude III

POVM element	E_c	Positive operator-valued measure element for case role c ; $P(c \mid \rho) = \text{Tr}(E_c \rho)$	§8
Prediction error	$\text{PE}(f)$	$\propto w_f \cdot \mu_{\text{predicted}} - \mu_{\text{observed}} $; case violation signal scaling with morphism weight	§7
Shannon entropy	H	Information-theoretic measure characterized as the unique derivation of a topological operad (Bradley)	§5
Topological operad	—	Operad with topological structure whose derivations connect magnitude to entropy	§5
Weight vector	$\mathbf{w}$	Solution to $Z\mathbf{w} = \mathbf{1}$; entries are the effective weights of each object	§5

Distributional Semantics and LLMs I

Table 4: Distributional semantics and LLM terminology.

Term	Symbol	Definition	First Use
Attention mechanism	—	Transformer component computing weighted relevance between token positions	§4
Attention weight	α_{ij}	Softmax-normalized score encoding contextual relevance of token j to token i	§4
BERT	—	Bidirectional Encoder Representations from Transformers; masked language model producing contextualized embeddings	§4
Contextualized word embedding	$\mathbf{z}_w^{(c)}$	Vector representation of word w that varies with linguistic context c	§4

Distributional Semantics and LLMs II

Compact closure map	$\varepsilon_N : N \otimes N \rightarrow \mathbb{R}$	Inner product implementing pregroup contraction in the vector space semantics	§4
Cosine similarity	$\cos(\mathbf{u}, \mathbf{v})$	Similarity measure $\frac{\mathbf{u} \cdot \mathbf{v}}{\ \mathbf{u}\ \ \mathbf{v}\ }$ between vectors	§5
Distributional Memory	—	Baroni–Lenci tensor-based framework structuring co-occurrence as (word, relation, word) triples	§4
DisCoCat	—	Distributional Compositional Categorical model; monoidal functor from pregroup grammars to vector spaces	§3
DisCoCirc	—	Discourse-level extension of DisCoCat with persistent entity wires	§4
Distributional hypothesis	—	The thesis that words occurring in similar contexts have similar meanings (Harris 1954, Firth 1957)	§4

Distributional Semantics and LLMs III

GloVe	—	Global Vectors for Word Representation; log-bilinear model of word co-occurrence statistics	§4
GPT	—	Generative Pre-trained Transformer; autoregressive language model	§4
Meaning functor	$F : \mathbf{Preg} \rightarrow \mathbf{FVect}$	Monoidal functor assigning vector spaces to types and linear maps to derivations	§4
Noun space	N	Vector space to which noun types n map under the meaning functor	§4
Parameterized- optic	$\mathbb{C}^k$	Categorical construction (Gavranović) modeling attention heads as functorial lenses	§9
Self- attention	$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$	Transformer operation computing scaled representations	§4
Sentence space	S	Vector space to which sentence type s maps under the meaning functor	§4

Distributional Semantics and LLMs IV

Sentence vector	$\overline{\mathbf{v}_S}$	Vector in S computed by tensoring and contracting word meanings via DisCoCat	§4
Static embedding	$\mathbf{v}_w$	Fixed vector representation of word w independent of context (e.g., Word2Vec, GloVe)	§4
Transformer	—	Neural architecture using self-attention and feed-forward layers for sequence processing	§4
Word2Vec	—	Neural embedding model (Mikolov et al. 2013) learning word vectors from local context windows	§4
Word vector	$\mathbf{v}_w \in \mathbb{R}^d$	d -dimensional real-valued vector encoding distributional properties of word w	§4

Distributional Active Inference (DAIF) I

Table 5: Distributional Active Inference (DAIF) terminology.

Term	Symbol	Definition	First Use
Return distribution	$Z(s, a)$	Full probability distribution over discounted cumulative rewards from state s under action a	§7c
Distributional Bellman operator	$\mathcal{T}^\pi$	Distributional analogue of the Bellman operator: $\mathcal{T}^\pi Z \stackrel{d}{=} R + \gamma Z'$	§7c
Push-forward measure	$S_\# \mathbb{P}$	Path-level push-forward of the trajectory measure (composed with decoder $f : \mathcal{S} \rightarrow \mathcal{X}$ in ??); generic push-forward written $T_\# \mu$	§7c
Discount factor	$\gamma \in (0, 1)$	Exponential time-discount in the distributional Bellman return	§7c

Distributional Active Inference (DAIF) II

Quantile level	$\tau \in [0, 1]$	Quantile index in QR-DQN and IQN; sampled uniformly at inference time	§7c
Quantile Huber loss	$\rho_{\tau}^{\kappa}(u)$	Loss function for quantile regression: interpolates L^1 and L^2 via threshold κ	§7c
Wasserstein distance	$W_p(P, Q)$	Wasserstein distance between return distributions; W_1 = area between CDFs	§7c
Bethe free energy	F_{Bethe}	Tractable approximation to variational free energy in belief propagation	§7c
Expected information gain	$\text{ElG}(o)$	$D_{\text{KL}}(q(\mathbf{c} \mid o) \parallel p(\mathbf{c}))$; epistemic value of observation o	§7c
Expected free energy	$G(\pi)$	Pragmatic + epistemic + risk cost of policy π ; minimised for action selection	§7c
Distributional prediction error	DPE	Precision-weighted Wasserstein mismatch: $w_f \cdot W_1(Z_{\text{pred}}, Z_{\text{obs}})$	§7c

Distributional Active Inference (DAIF) III

ERP amplitude profile	ERPP Data	Default: class holding N400/P600 amplitudes, peak latencies, and time-series waveforms for each case role	§7c
Return entropy	$H[Z]$	Shannon entropy of the return distribution: $-\sum_i p_i \log p_i$	§7c
Quantile calibration error	CE	Mean absolute deviation between nominal quantile levels and empirical coverage	§7c
IQN risk distortion	$\psi_{\text{IQN}}(\tau)$	Maps quantile levels τ ; piecewise formulas (neutral, power, tail, CVaR) match <code>implicit_quantile_network_update()</code> and the §7c table; distinct from β_{risk} in $G(\pi)$	§7c
IQN curvature	η_{IQN}	Default 0.71 in code; not the compact-closure cap η_{cc}	§7c
CVaR scale	α_{CVaR}	IQN mode uses $\psi_{\text{IQN}}(\tau) = \tau \cdot \alpha_{\text{CVaR}}$; default 0.25 in code (distinct from <code>softmax(α_{CVaR})</code>)	§7c

Distributional Active Inference (DAIF) IV

Risk sensitivity (EFE)	β_{risk}	Non-negative coefficient on risk(π) in $G(\pi)$; $\beta_{\text{risk}} = 0$ recovers standard EFE	§7c
Inverse temperature (policy)	α_{pol}	Softmax sharpness over policies in $P(\pi)$	§7c
Belief precision (VMP)	$\Lambda_{\text{post}}, \Lambda_{\text{prior}}, \Lambda_{\text{lik}}$	Posterior, prior, and likelihood precision matrices; $\Delta\Lambda$ is their positive update magnitude	§7c
Violation severity	$S_{\text{violation}}$	$\in \{0, 0.5, 1.0\}$ in ERP decomposition	§7c
TD error (quantile)	δ_{ij}	Temporal-difference residual in QR-DQN (??)	§7c
Quantile count (current)	N	Number of current-network quantile levels in QR-DQN (Eq. ??); equal to the length of <code>current_quantiles</code> passed to <code>quantile_td_update()</code>	§7c

Distributional Active Inference (DAIF) V

Quantile count (target)	N'	Number of target-network quantile samples in QR-DQN (Eq. ??)	§7c
Huber threshold	κ	Cut-point between quadratic and linear regimes of the Huber loss $L_{\kappa}(u)$ (Eq. ??); default $\kappa = 1$ in <code>quantile_td_update()</code>	§7c
Atom count (C51)	N_{atoms}	Number of support atoms z_i in the C51 categorical representation (Eq. ??); default $N_{\text{atoms}} = 51$ in <code>categorical_return_distribution()</code>	§7c
Support bounds (C51)	$V_{\min}, V_{\max}$	Lower and upper endpoints of the atomic support spanned by $\{z_i\}$ in the C51 representation	§7c

Distributional Active Inference (DAIF) VI

Bin count (en- tropy)	N_{bins}	Number of equal-width bins used to discretise the quantile-parameterised return when computing $H[Z]$ (Eq. ??); default $N_{\text{bins}} = 50$ in <code>return_distribution_entropy()</code>	§7c
--------------------------------	-------------------	--	-----

Active Inference and Cognitive Models I

Table 6: Active inference and cognitive modeling terminology.

Term	Symbol	Definition	First Use
Active inference	—	Framework in which perception and action are unified as variational free energy minimization	§7
Active sampling	—	Agent's selection of actions to confirm or update case assignments via sensory evidence	§7
Belief updating	—	Bayesian posterior computation: revising generative model parameters given new observations	§7
CEREBRUM	—	Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling; treats AI models as case-bearing entities	§7

Active Inference and Cognitive Models II

Distributional- Active Infer- ence (DAIF)	—	Extension replacing scalar value summaries with full return distributions in active inference	§7
Distributional- RL	—	Reinforcement learning operating on full return distributions rather than scalar expected values	§7
Free energy	F	Variational bound on surprisal; $F = D_{KL}[q(\theta) \parallel p(\theta \mid o)] - \ln p(o)$	§7
Free energy principle (FEP)	—	The principle that self-organizing systems maintain themselves by minimizing surprisal	§7
Garden- path reanaly- sis	—	Restructuring of case assignments when incoming evidence contradicts the current parse	§7

Active Inference and Cognitive Models III

Generative model	$p(o, s)$	Joint probability model over observations o and hidden states s	§7
Markov blanket	—	Statistical boundary separating internal states from external environment; defines agent boundary	§7
N400	—	Event-related brain potential peaking ~400ms post-stimulus; indexes semantic prediction error	§7
P600	—	Event-related brain potential peaking ~600ms post-stimulus; indexes syntactic prediction error	§7
Perceptual inference	—	Updating internal beliefs to better predict current observations (reduce prediction error)	§7
Precision weighting	—	Weighting of prediction errors by inverse variance; enriched morphism weights serve this role	§7

Active Inference and Cognitive Models IV

Prediction error	ε	Difference between predicted and observed sensory input; drives belief updating	§7
ROSE model	—	Murphy's Representation–Operation–Structure–Encoding architecture; cross-frequency phase-amplitude coupling (PAC) linking biolinguistic syntax (slow oscillatory phase) to neuropragmatic inference (fast gamma); the neural-timescale bridge of Pillar 6	§1, §7c
S-HAI	—	Schema-Based Hierarchical Active Inference; dual-level POMDP connecting abstract relational schemas (Level 2: case diagram structure) to sensorimotor surface parsing (Level 1)	§7
Push-forward (general)	$T_{\#}\mu$	Image of a measure μ under a measurable map T ; DAIF-specific notation in 24	§7

Active Inference and Cognitive Models V

Situation semantics	—	Framework representing meaning as relations between situations (Barwise and Perry 1983)	§7
Surprisal	$-\ln p(x)$	Negative log-probability of an observation under the generative model	§7
Variational free energy	$F[q]$	Functional upper bound on surprisal minimized by approximate posterior q	§7

Quantum and Topological Terms I

Table 7: Quantum and topological terminology.

Term	Symbol	Definition	First Use
Amplituhedron	—	Positive geometry encoding scattering amplitudes; connected to TQNN execution traces	§8
Barren plateau	—	Vanishing gradient phenomenon in PQC training; gradients decay exponentially in system size for certain ansätze	§4b
$\dagger$ -compact category	—	See $\dagger$ -compact closed category in Category Theory	§8
CPTP map	—	Completely Positive Trace-Preserving map; quantum channel between Hilbert spaces	§8
Density operator	ρ	Positive semidefinite trace-one operator encoding a quantum (or semantic) state	§8

Quantum and Topological Terms II

Execution trace	—	Record of operations in a quantum computation; connects TQNNs to amplituhedra	§8
Generalized flow	—	Graph-theoretic property ensuring deterministic circuit extraction from ZX-diagrams	§8
Hadamard box	H	ZX-calculus element implementing the Hadamard gate; converts between Z and X bases	§8
Holographic screen	—	Information boundary between interacting quantum systems carrying a qubit array	§8
Parameterized quantum circuit (PQC)	—	Quantum circuit with trainable angle parameters; the computational substrate of QNLP and case-category implementations	§4b
IQP ansatz	—	Instantaneous Quantum Polynomial-time circuit: default lambeq PQC ansatz for noun and verb boxes	§4b

Quantum and Topological Terms III

Sim4 ansatz	—	Strongly entangling layer PQC ansatz used for discourse-level lambeq Gen II circuits	§4c
Pointer state	—	Preferred quantum state selected by a QRF; determines the measurement basis	§8
Quantum context- uality	—	Quantum correlations that reduce cohomological obstructions to semantic alignment	§8
Quantum discord	—	Quantum correlation measure equal to integrated semantic information in sheaf framework	§8
Quantum key dis- tribution (QKD)	—	Protocol providing information-theoretic security for quantum communication channels	§9
Quantum refer- ence frame (QRF)	—	Observer-relative frame selecting pointer states and inducing decoherence	§8

Quantum and Topological Terms IV

Semantic Hilbert space	H_v	The finite-dimensional semantic Hilbert space carried at vertex v in a quantum semantic sheaf	§8b
Quantum semantic sheaf	(H, F, τ)	Triples of Hilbert spaces, CPTP channels, and density operators over a communication graph	§8
Reshetikhin—Turaev invariant	—	Topological invariant assigning to a ribbon graph a linear map via TQFT	§8
Sheaf cohomology	$H^n(\mathcal{F})$	Cohomological obstruction classes governing alignment in a quantum semantic sheaf	§8
Spin-network	—	Graph with edges labeled by representations and vertices by intertwiners; TQFT data	§8
Spider	—	Elementary ZX-diagram node (Z-spider or X-spider) representing a quantum operation	§8

Quantum and Topological Terms V

TQFT	—	Topological Quantum Field Theory; functor from cobordisms to Hilbert spaces	§8
TQNN	—	Topological Quantum Neural Network; QNN reformulated via spin-networks and TQFT	§8
Turaev–Viro invariant	—	State-sum TQFT invariant; implements quantum error-correcting codes in TQNNs	§8
ZX-calculus	—	Graphical language for quantum circuits using Z-spiders, X-spiders, and Hadamard boxes	§8
ZX-diagram	—	String diagram in a $\dagger$ -compact closed category representing a quantum process	§8
ZX rewrite rule	—	Graph-theoretic transformation preserving the semantics (linear map) of a ZX-diagram	§8

Logical and Type-Theoretic Terms I

Table 8: Logical and type-theoretic terminology.

Term	Symbol	Definition	First Use
β -reduction	—	Computational reduction step in λ -calculus; corresponds to cut elimination in proofs	§3
Church–Rosser property	—	Confluence of β -reduction: all reduction sequences converge to the same normal form	§3
Curry–Howard isomorphism	—	Correspondence between proofs and programs, propositions and types	§3
Cut elimination	—	Proof normalization procedure removing intermediate lemmas; corresponds to β -reduction	§3

Logical and Type-Theoretic Terms II

Graded type theory	—	Extension of type theory tracking effects (e.g., evidentiality) via graded modalities	§3
Lambek calculus	—	Non-commutative intuitionistic linear logic for syntactic type assignment	§3
Left residual	$A \backslash B$	Type of an expression that, given A to the left, produces B	§3
Residuation law	$A \otimes B \leq C \iff A \leq C/B$	Fundamental axiom connecting the three connectives of the Lambek calculus	§3
Right residual	B/A	Type of an expression that, given A to the right, produces B	§3

AI and Communication Protocols I

Table 9: AI and communication protocol terminology.

Term	Symbol	Definition	First Use
A2A Protocol	—	Google's Agent-to-Agent protocol for cross-framework agent communication via HTTP/JSON-RPC	§9
ACP	—	Agent Communication Protocol; standardizes messaging formats across agents, apps, and humans	§9
ANP	—	Agent Network Protocol; three-layer architecture for trusted distributed agent interaction	§9
Categorical deep learning	—	Deep learning approached through the lens of category theory (Gavranović et al.)	§9

AI and Communication Protocols II

Double Categorical Systems Theory (DCST)	—	Framework using 2-categories (horizontal + vertical composition) for explainable autonomous AI	§9
Functorial encryption	—	Semantic cryptography: applying a secret functor to map plaintext categories into ciphertext categories	§9
lambeq	—	Quantum Natural Language Processing pipeline compiling DisCoCat diagrams to quantum circuits	§9
MCP	—	Model Context Protocol; standardizes how AI agents access external tools and data sources	§9
Parameterized optics / lenses	—	Categorical constructions modeling neural network components (Gavranović); attention heads as optics	§9

AI and Communication Protocols III

QNLP	—	Quantum Natural Language Processing; quantum computation on DisCoCat sentence diagrams	§9
Role variables	$X_{\text{NOM}}, X_{\text{ACC}}$	Authentic components structured by case roles in networked LLM contexts (e.g., active requester policy)	§9
Semantic cryptog- raphy	—	Encrypting compositional meaning structures (functorial encryption, diagram obfuscation, weight masking)	§9
Protocol category	$\mathcal{C}_{\text{protocol}}$	Category whose morphisms are licensed interaction steps in the case-theoretic firewall (??)	§9b
Adversarial mor- phism	ϕ	Illicit map attempting case-role promotion (e.g. ACC→NOM) in injection attacks	§9b
Access Collapse	—	Catastrophic boundary failure where adversarial text traverses from passive data (ACC) to active instruction (NOM) without structural constraint	§9b

AI and Communication Protocols IV

Case-theoretic firewall	—	Type-checking system enforcing licensed morphism constraints at agent communication boundaries; detects illicit role promotions in $\text{Mor}(\mathcal{C}_{\text{protocol}})$	§9b
Prompt injection	—	Attack illicitly promoting user-supplied data from ACC (patient) to NOM (commanding agent); structurally a type violation in the interaction grammar	§9b
Symbolic Isolation	—	Property ensuring ACC-typed data wires cannot fuse with NOM-typed command wires; enforced by the non-cartesian fragment of the monoidal structure	§9b

Diagrammatic Reasoning I

Table 10: Diagrammatic reasoning terminology.

Term	Symbol	Definition	First Use
Categorical complexity	$\kappa(D)$	Complexity metric for diagram D derived from box counts, cup counts, and swap depths	§4b
Cognitively privileged representation	—	Representation format that leverages perceptual and spatial cognition for inference	§1
Diagram depth	—	Length of the longest input-to-output path through boxes; measures derivational complexity	§4
Existential graphs	—	Peirce's graphical logic system conducting first-order logic entirely diagrammatically	§7

Diagrammatic Reasoning II

Free ride	—	Shimojima's term: information extracted from a diagram without explicit inference steps	§1
Hybrid reasoning	—	Giardino's term: reasoning combining perceptual pattern recognition with theoretical knowledge	§1
Inferential instrument	—	Manders's term: a diagram whose spatial properties encode proof-relevant information	§6
Joyal–Street theorem	—	Soundness and completeness of string-diagrammatic reasoning for monoidal categories	§3
Normal form	D_{nf}	Canonical form of a diagram obtained by rewriting; unique up to the axioms	§4

Diagrammatic Reasoning III

Role colours (figures)	CASE_COLORS → display colour for generated figures; single source in src/visualization/styles.py	§2– §4, Ap- pendix A
------------------------------	--	----------------------------------

Notation Conventions I

Table 11: General mathematical notation and manuscript conventions.

Convention	Meaning
$\mathcal{C}, \mathcal{D}$	Categories
$\mathcal{V}$	Enrichment base (monoidal category, typically $([0, 1], \cdot, 1)$)
$\mathcal{U}$	Universal (maximal) case category
$\mathcal{L}$	Language-specific case category
$\mathcal{E}_{\mathbb{T}}$	Classifying topos of theory $\mathbb{T}$
f, g, h	Morphisms
F, G	Functors
α, β	Natural transformations; functorial vertical composition (componentwise along objects) is standard (§2). In DAIF, α_{pol} is policy temperature; α_{CVaR} is CVaR scale; β_{risk} is EFE risk weight; see §7c
n, s	Basic pregroup types: noun, sentence

Notation Conventions II

n^l, n^r	Left and right adjoints of type n
$n_{\text{NOM}}, n_{\text{ACC}}$	Case-subscripted noun types (e.g., nominative noun, accusative noun)
N, S	Noun space and sentence space under the meaning functor
$N_{\text{NOM}}, N_{\text{ACC}}$	Case-specific vector subspaces in case-enriched DisCoCat
$\overrightarrow{\text{word}}$	Word vector (column vector in noun space N)
$\overleftrightarrow{\text{verb}}$	Verb tensor (element of $N \otimes S \otimes N$ for transitive verbs)
Preg	Category of pregroup types and reductions
FVect	Category of finite-dimensional vector spaces and linear maps
FdVect	
FHilb	Category of finite-dimensional Hilbert spaces and linear maps
Qubit	Category of qubit systems with tensor product structure
Set	Category of sets and functions

Notation Conventions III

Ent , Case	Categories for Entities and Case Roles respectively in DisCoCirc discourse extensions
$\otimes$	Tensor product (monoidal product, type concatenation)
$\circ$	Composition of morphisms
$\Rightarrow$	Natural transformation between functors
$\simeq$	Categorical equivalence
$\leq$	Preorder relation on types (derivability)
γ	Discount factor in distributional RL return computation
$w \in [0, 1]$	Enriched morphism weight (proto-role satisfaction degree)
ε	Sensory prediction error (active inference); ε_n also denotes the compact-closure counit (cup) on type n
η_{cc}	Compact closure unit (cap); IQN curvature uses η_{IQN} instead
ϵ	Small numerical floor (e.g. KL stabiliser, VMP convergence threshold); not the counit
ρ	Density operator (quantum state)

Notation Conventions IV

[@key] Parenthetical citation

[-@key] Suppress-author citation

\autoref LaTeX/Markdown automatic cross-reference to a
(with labeled target

{sec:...}

/

{eq:...}

/

{fig:...})

$Z(s, a)$ Return distribution (DAIF); full distributional
representation of returns

$G(\pi)$ Expected free energy of policy (DAIF active inference)

τ Quantile level in QR-DQN / IQN

$\psi_{\text{IQN}}(\tau)$ IQN risk distortion of quantile levels

β_{risk} Risk-sensitivity coefficient in $G(\pi)$

$\alpha_{\text{pol}} ; \alpha_{\text{CVaR}}$ Policy softmax temperature; CVaR tail level (IQN)

w_f, w_k, w_c Enriched morphism / role weights (precision on PE and
VMP)

Notation Conventions V

DPE	Distributional prediction error; precision-weighted Wasserstein mismatch
$H[Z]$	Return distribution entropy
